University
Mohammed VI
Polytechnic

College of
Computing

Lab 5: Normalization

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering
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Team Information

Member 1	BENYASSINE Rayane
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Member 3	Name: BITAR Driss
Repository Link	`https://github.com/Adam-Ait-Lahcen/Base-de-donn-es`

1 Introduction

This lab focuses on the principles of database normalization, specifically evaluating relational schemas against Boyce-Codd Normal Form (BCNF). It explores the use of functional dependencies to identify redundancies, resolve update anomalies through decomposition, and mathematically verify that these structural refinements are both lossless and dependency preserving.

2 Team Roles

The work was divided between the three members of the team. Driss BITAR and Adam AIT LAHCEN collaborated on Task 1, while Rayane BENYASSINE was responsible for writing the report and completing Task 2. However, these initial responsibilities do not mean that the final answers can be attributed exclusively to one or two members. After preparing the first versions, all three members worked together to modify, correct, and validate the answers. Therefore, the final solutions reflect the collective work of the entire team.

3 Key Challenges and Their Resolution

Overall, the lab was relatively simple, and we did not encounter any major difficulties. Most of the work consisted of applying the concepts of BCNF and 3NF to the given relations, identifying the functional dependencies that violate these normal forms, and performing the appropriate decompositions. One point we discussed was whether we needed to verify all possible keys and their subsets before concluding that we had found the candidate keys, or whether we could rely on the cases where the result was clear. We considered the importance of checking minimality and ensuring that no other candidate key existed. In the final answers, we focused on the relevant functional dependencies and the necessary checks to justify our conclusions. Another point concerned the normal forms below 3NF. Since we had not covered 2NF during the course, we did not test for violations of 2NF. Consequently, when a relation violated 3NF, we checked whether it satisfied 1NF and concluded that 1NF was the strongest normal form established by our analysis. This allowed us to remain within the concepts covered in the course. Apart from these points, we did not have significant disagreements, and the corrections were made collectively.

4 Justification of Conversion Choices

For Task 1, we concluded that none of the 31 relations from the Lab 4 solution violated BCNF, considering the functional dependencies stated in that solution. The justification was that, for each non-trivial functional dependency, its determinant was a candidate key or a superkey. For relations containing only key attributes, there were no non-trivial functional dependencies that could violate BCNF. Therefore, no decomposition was necessary for Task 1. For Task 2, we examined five relations and identified the functional dependencies that violated BCNF. The decompositions were chosen according to these violating dependencies. The general approach was to separate the attributes determined by a determinant that was not a superkey into a new relation, while retaining

the attributes needed to preserve the original relationship. For each decomposition, we checked whether the resulting relations satisfied BCNF. We also verified lossless join and dependency preservation. These checks justified our choices because they ensured that the decompositions satisfied the required normal form without losing information or making the given functional dependencies impossible to enforce locally. In the cases where 3NF was violated, we concluded that the relation satisfied 1NF, which was the strongest normal form we established, given that 2NF had not been covered in the course.

5 Assumptions Made

We assumed that the relations satisfied 1NF, as the attributes were considered atomic. We did not investigate possible violations of 2NF because this normal form had been covered during the course.